Neural Networks



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Learning Objectives

- Motivation
- What is a Neuron?
- Components of an Artificial Neural Network (ANN) or NN or MLP
- Training of a NN
    - Forward Pass
    - Backward Pass
    - Update of weights
- Binary and Multiclass Classification Neural Networks

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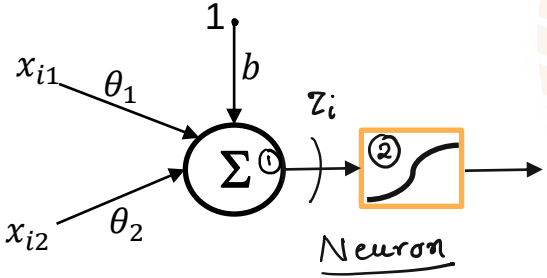
Motivation

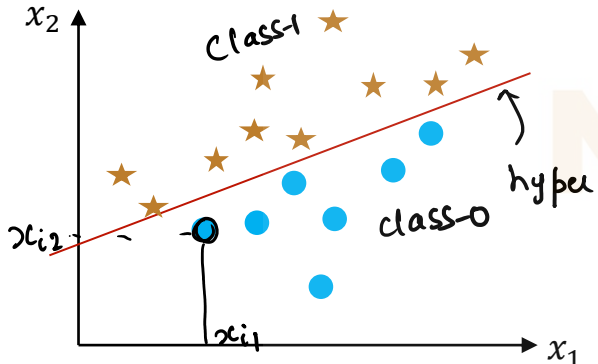
- **Neural Networks (NN)**: A type of ML algorithm that are inspired by the structure and function of the human brain. The elements of a NN are neurons and its interconnections. The interconnections have weights that are called as the learnable parameters which are required to be learned through the process of training.

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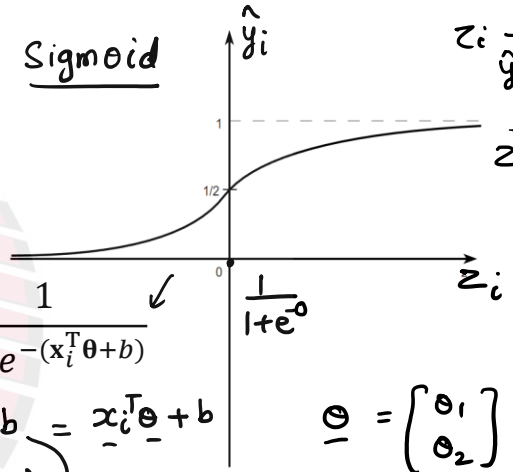
A Neuron

- **A Neuron**: Contains two operations
  - ① a weighted linear combiner
  - ② A non-linear activation function





Sigmoid



$z_i \to \infty$, $\hat{y}_i = \frac{1}{1+e^{-\infty}} = 1$

$z_i \to -\infty$, $\hat{y}_i = \frac{1}{1+e^{\infty}} = \frac{1}{\infty} = 0$

$$\hat{y}_i = p(y_i = 1/\mathbf{x}_i, \boldsymbol{\theta}, b) = \frac{1}{1+e^{-(\mathbf{x}_i^T\boldsymbol{\theta}+b)}}$$

① $z_i = x_{i1}\theta_1 + x_{i2}\theta_2 + b = \underline{x}_i^T\underline{\theta} + b$

$\underline{\theta} = \begin{bmatrix}\theta_1\\ \theta_2\end{bmatrix}$

i^{th} sample

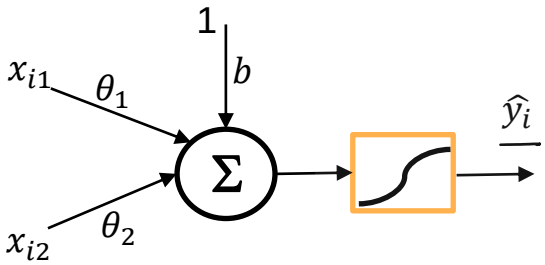
learnable parameters $\langle \underline{x}_i, \underline{\theta} \rangle = \underline{x}_i^T\underline{\theta}$

features

$\underline{x}_i = \begin{bmatrix}x_{i1}\\ x_{i2}\end{bmatrix}$

② $\hat{y}_i = \frac{1}{1+e^{-z_i}}$

A Neuron



$$\hat{y}_i = p(y_i = 1/\mathbf{x}_i, \boldsymbol{\theta}, b) = \frac{1}{1 + e^{-(\mathbf{x}_i^T \boldsymbol{\theta} + b)}}$$

Traing (x_i, y_i)

Ground truth (GT)

The binary cross entropy loss function = $\mathcal{J}(\theta)$ = -log Likelihood function = $-\ln l(\boldsymbol{\theta})$

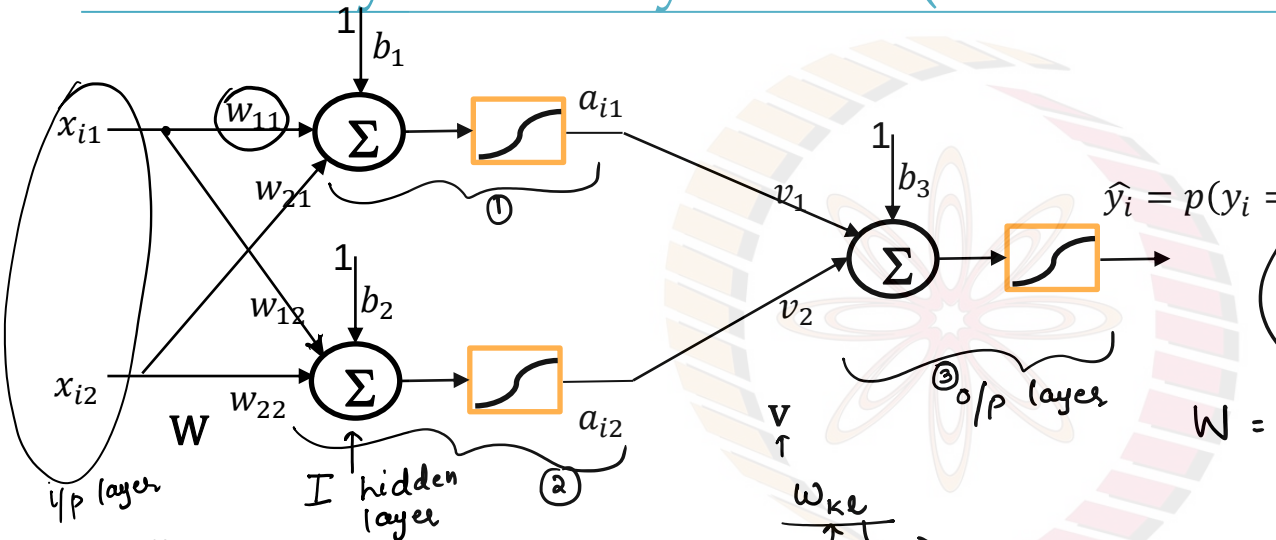
$$\mathcal{J}_i(\theta) = -\ln l_i(\boldsymbol{\theta}) = -y_i \ln p(y_i = 1/\mathbf{x}_i, \boldsymbol{\theta}, b) - (1 - y_i) \ln p(y_i = 0/\mathbf{x}_i, \boldsymbol{\theta}, b)$$

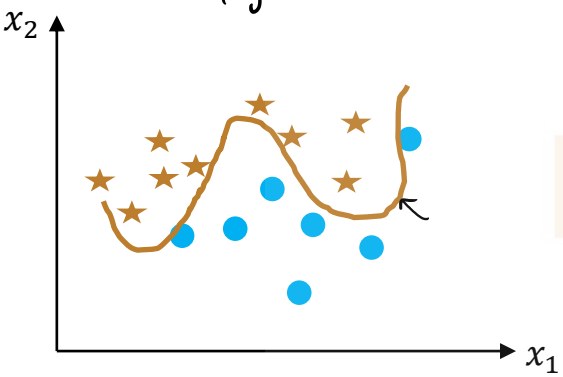
$$= -y_i \ln \hat{y}_i - (1 - y_i) \ln(1 - \hat{y}_i) = \begin{cases} -\ln \hat{y}_i & \text{when } y_i = 1 \\ -\ln(1 - \hat{y}_i) & \text{when } y_i = 0 \end{cases}$$

For n no. of training samples: Total BCE loss is given by

$$\mathcal{J}(\theta) = -\ln l(\boldsymbol{\theta}) = -\sum_{i=1}^{n} [y_i \ln p(y_i = 1/\mathbf{x}_i, \boldsymbol{\theta}, b) + (1 - y_i) \ln p(y_i = 0/\mathbf{x}_i, \boldsymbol{\theta}, b)]$$

A two layer binary ANN (MLP classifier)





Note: $\boldsymbol{\theta}$ denotes all the learnable parameters.

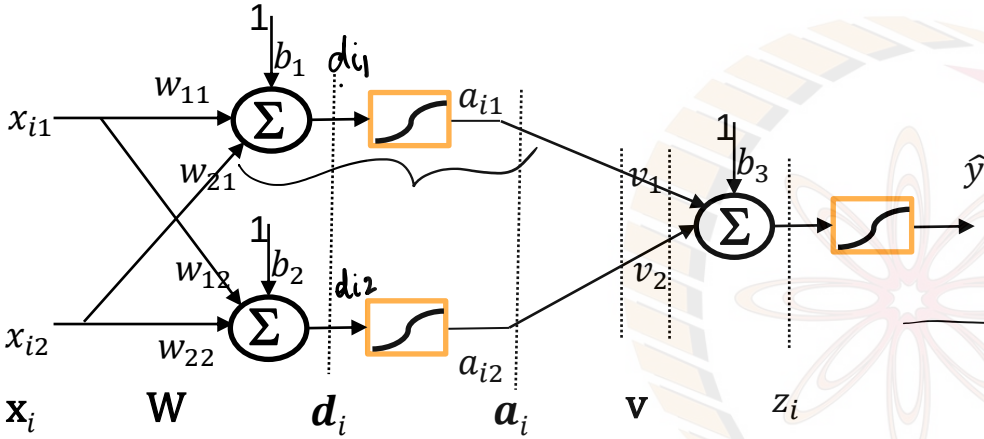
learnable parameters

$$\begin{pmatrix} w_{11} & w_{12} & w_{21} & w_{22} & b_1 & b_2 \\ v_1 & v_2 & b_3 \end{pmatrix}$$

$$W = \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \end{bmatrix}$$

$$\underline{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

A two layer binary ANN (MLP classifier)



$$W = \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \end{bmatrix} = \begin{bmatrix} \underline{w}_1 & \underline{w}_2 \\ \downarrow & \downarrow \end{bmatrix}$$

$$a_{i1} = \frac{1}{1+e^{-d_{i1}}} = \frac{1}{1+e^{-(\underline{w}_1^T \underline{x}_i + b_1)}}$$

$$a_{i2} = \frac{1}{1+e^{-d_{i2}}} = \frac{1}{1+e^{-(\underline{w}_2^T \underline{x}_i + b_2)}}$$

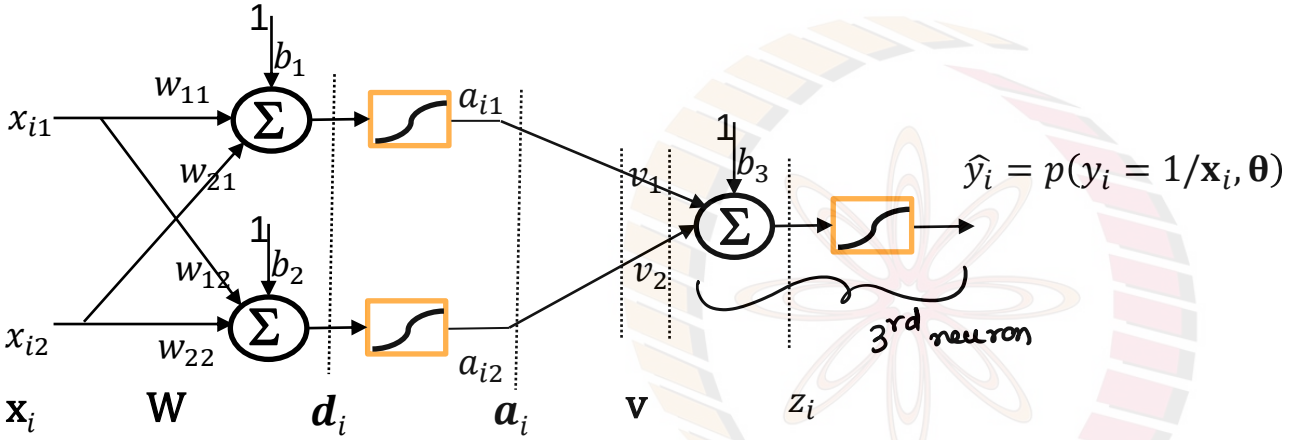
$$\underline{a}_i = \begin{bmatrix} a_{i1} \\ a_{i2} \end{bmatrix}$$

$$\underline{d}_i = W^T \underline{x}_i + \underline{b}$$

$$\begin{bmatrix} d_{i1} \\ d_{i2} \end{bmatrix} = \begin{bmatrix} w_{11} & w_{21} \\ w_{12} & w_{22} \end{bmatrix} \begin{bmatrix} x_{i1} \\ x_{i2} \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$\begin{bmatrix} d_{i1} \\ d_{i2} \end{bmatrix} = \begin{bmatrix} w_{11} x_{i1} + w_{21} x_{i2} + b_1 \\ w_{12} x_{i1} + w_{22} x_{i2} + b_2 \end{bmatrix} = \begin{bmatrix} \underline{w}_1^T \underline{x}_i + b_1 \\ \underline{w}_2^T \underline{x}_i + b_2 \end{bmatrix}$$

A two layer binary ANN (MLP classifier)



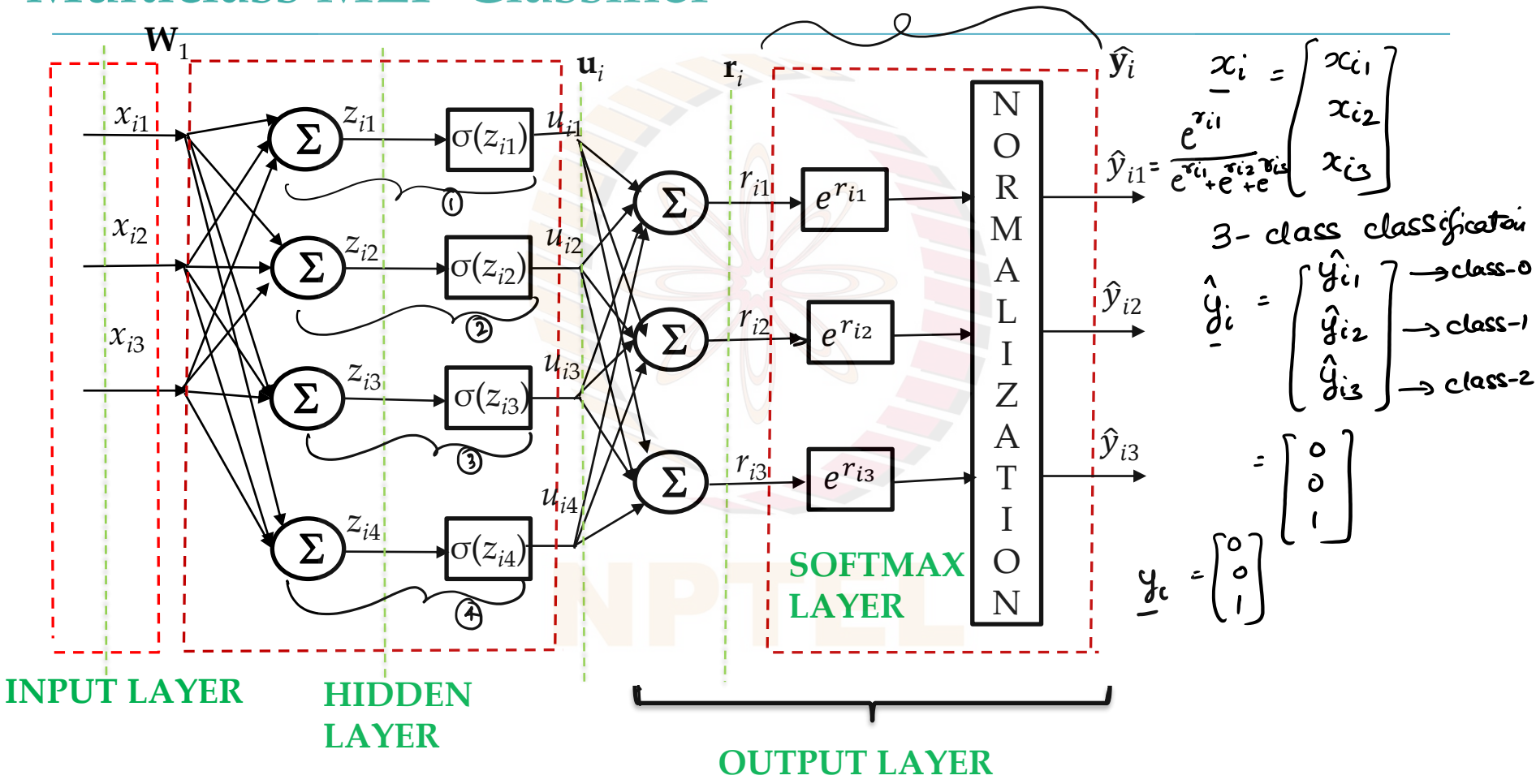
$$z_i = a_{i1} v_1 + a_{i2} v_2 + b_3$$

$$= \underline{a}_i^T \underline{v} + b_3$$

$$= \underline{v}^T \underline{a}_i + b_3$$

$$\hat{y}_i = \frac{1}{1 + e^{-z_i}} = \frac{1}{1 + e^{-(\underline{v}^T \underline{a}_i + b_3)}}$$

Multiclass MLP Classifier



Training of a Neural Network

Training of a NN consists of the following steps:

- **Step-1: Forward Propagation** of the training data
- **Step-2: Computation of the loss** or the objective function as a function of error between the true label and the predicted label
- **Step-3: Backpropagate the error gradients** with respect to the learnable parameters (**weights and biases**)
- **Step-4**: Complete Steps 1-3 for the entire training data once (constituting an **epoch**).
- **Step-5**: Repeat Steps 1-4 for a number of epochs until the **loss is minimised or an evaluation metric is** optimised. **At this stage, weights and bias values** are considered to be optimum and hence, the model is considered trained.
- **Step-6**: Take **prediction on the new data** using the trained NN.

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Summary

In this module,

- We looked at the motivation for studying a neural network
- We studied the concept of a Neuron in the context of ML.
- We looked at the components of an Artificial Neural Network (ANN) or NN or MLP
- We have noted that the training of a NN involves three steps:
  - Forward Pass
  - Backward Pass
  - Update of weights
- We studied the neural networks for binary and multiclass classification tasks.

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